CompTIA Server+

Securing a Physical Server Infrastructure

Introduction

Server+
Physical Security
Motion Sensors
Bollards
Environmental Control
Interior Access Control

Welcome to the **Securing a Physical Server Infrastructure** Practice Lab. In this module, you will be provided with information on how server infrastructure can be secured physically.

Servers have become so vital to our organizations. They provide services that allow us to function and provide security for our users' data. Making sure that these servers are not easily accessed is crucial because they have become mission critical. In this module, exterior and interior physical security measures that can be implemented for servers will be discussed.

Learning Outcomes

In this module, you will complete the following exercises:

- Exercise 1 - Exterior Physical Security
- Exercise 2 - Interior Physical Security

After completing this module, you should have further knowledge of:

- Fences and Security Walls

- Motion Sensors
- Lighting
- Gated Lots
- Security Guards
- Security Cameras
- Signal Blocking
- Reflective Glass
- Data Center Camouflage
- Bollards
- Exterior Door Locks
- Interior Access Control
- Environmental Controls

Exam Objectives

The following exam objectives are covered in this module:

3.2 Summarize physical security concepts

- Physical access controls
- Environmental controls

Lab Duration

It will take approximately **1 hour** to complete this lab.

Help and Support

For more information on using Practice Labs, please see our **Help and Support** page. You can also raise a technical support ticket from this page.

Click **Next** to view the Lab topology used in this module.

Lab Topology

This module contains supporting materials for Server+ (SK0-005).

Click **Next** to proceed to the first exercise.

Exercise 1 - Exterior Physical Security

Organizations need to have strong physical security. It is essential to ensure the safety of the valuable information that is frequently stored on the computers and servers that are used only for business, and confidentiality and integrity are preserved. This procedure begins outside the building and, depending on the sensitivity of the data that has to be protected, may require varying degrees of building access control and physical protection.

In this exercise, you will learn about physical security measures such as fences, security walls, motion sensors, lighting, and other precautions that can be put in place before the front or rear doors to keep the premises secure.

Learning Outcomes

After completing this exercise, you should have further knowledge of:

- Fences and Security Walls
- Motion Sensors
- Lighting
- Gated Lots
- Security Guards
- Security Cameras
- Signal Blocking
- Reflective Glass
- Data Center Camouflage
- Bollards
- Exterior Door Locks

Your Devices

This exercise contains supporting materials for Server+ (SK0-005).

Fences and Security Walls

Depending on the facility, fences and security walls may be used to help secure the facility or specific areas of the facility. The type and style of fences or border walls will depend on the type of business being done, the level of security needed and the budget. High towering fences with razor wire might not be necessary and appealing to areas where customers access the building. Areas like data centers might be more applicable for a fence like that. Not all fences are created equal, and they come in a variety of materials and styles. Below are some common styles of fencing and security walls.

Chain Link Fencing

Chain link fencing is a popular choice for those looking to fortify their perimeter and prevent illegal entry. This type is long-lasting, corrosion-resistant, and simple to install. It is also open, so you can see what is on the other side reducing blind spots. These fences can create spaces for footholds so they can be scalable. A fence topper like razor wire might be necessary to deter someone from climbing.

Figure 1.1: Displaying a chain link fence.

Welded Mesh

Welded mesh is a steel fence made up of high-strength welds between wires, making it more durable than other types of material. These fences can create spaces for footholds and can be scalable. A fence topper like razor wire might be necessary to deter someone from climbing.

Figure 1.2: Displaying metal fence made of welded mesh.

Vertical Bar Fencing

Vertical bar fencing is a strong and long-lasting type of security fence. Vertical bars pass through horizontal beams, which securely hold the posts together, making them difficult to bend. Due to the pleasing aesthetics, this fence is commonly used in commercial and recreational environments and parking lots.

Electric Fences

Electric fences are a safe and effective barrier for humans, wildlife, and intruders. They make perimeter security far more secure and easier to manage. An electric fence is normally made up of an energizer, conductor, posts, and insulators. When an animal or person contacts the fence, the energizer generates a current pulse at a predetermined frequency, causing them to suffer a shock. There are lethal and non-lethal versions depending on the need.

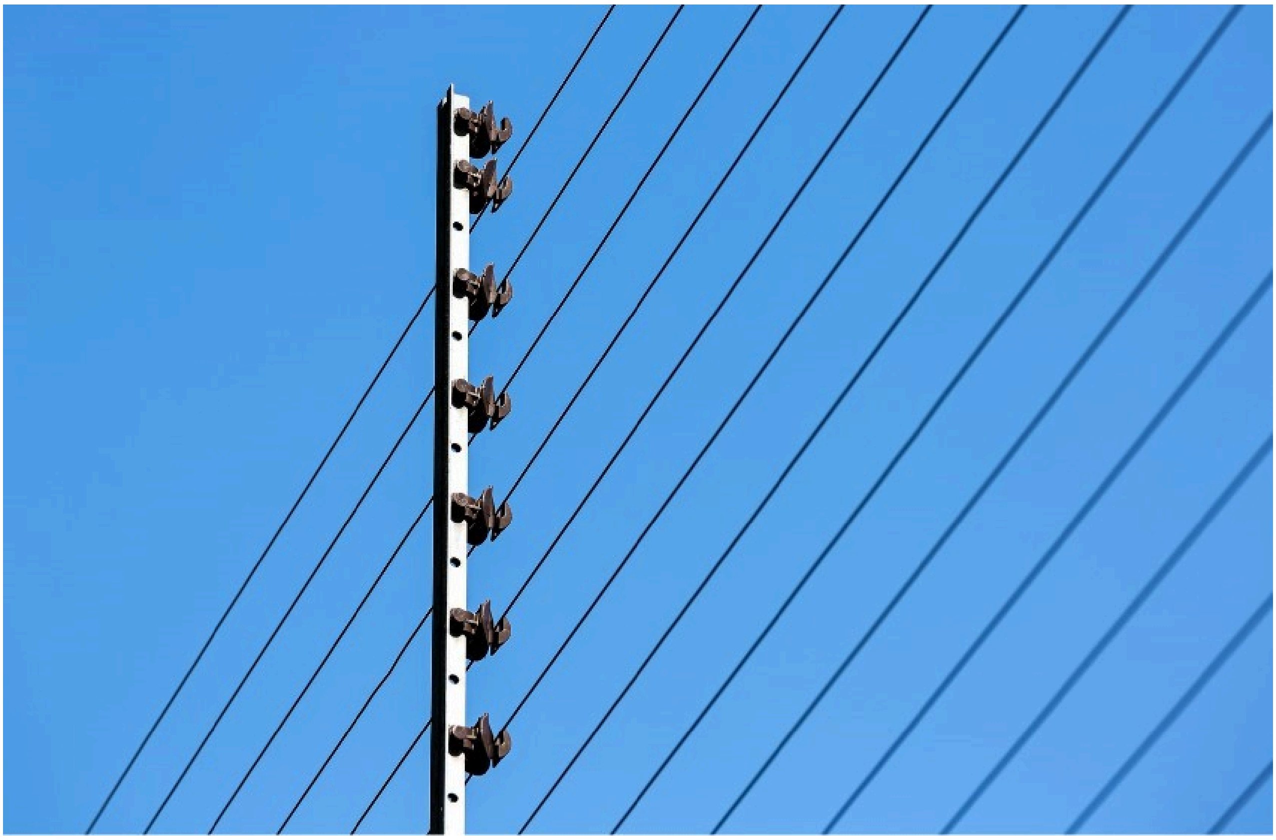

Figure 1.3: Displaying an electric fence.

Border Walls

Border walls can be created to help provide security. These walls will provide a barrier with entry and exit points for access control. The walls can be constructed of simple cinder blocks or bricks but can also be more elaborate with stone décor. Border walls can be a great form of protection, but there are some downsides. The wall likely will not allow us to see the other side, which can create blind spots in the security. The cost to build one of these walls is much higher than fencing. The wall can block water, and flooding can occur, so proper measures will need to be taken to ensure proper irrigation.

One key is to ensure you buy security-grade fencing and not just demarcation fencing. Security grade fencing will be designed and tested to provide a higher level of protection and durability. Some common things to look for in a fence would be:

- High grade metals used
- Anti-climb design
- Strong welds
- Open design to see through and take away blind spots
- Topped with a deterrent
- Lighting
- Sensors

Motion Sensors

Motion sensors can be added to the perimeter fence to create a deterrent or to alert us to a possible intrusion. They can be attached to activate lighting, cameras, and silent or active alarms that can alert us to an intrusion. There are three main types of motion sensors:

- **Passive Infrared (PIR) Sensors** can detect infrared radiation, allowing heat and motion detection. Many uses have been found for these sensors. They can be used for the proximity sensor, alarms, counting, thermometer, and detecting humans or animals. A common use for infrared sensors is to be attached to a light source, and when the thermal sensor is activated, the light switch is activated. After a period of no activity, the switch will automatically shut off. There are some common benefits to using infrared, like low power consumption, smaller and more affordable, highly accurate, and is not affected by the light
- The **Microwave Sensor** is active and constantly sends out electromagnetic frequencies. When these frequencies bounce off objects, their speed and direction are captured. It is mostly used in radar guns. Compared to PIR sensors, it has more range, accuracy, penetration, and overall capabilities. Because this is active monitoring, it will require higher amounts of electricity and is more costly.
- **Dual Tech/Hybrid Sensors** will use multiple methods to increase the accuracy and efficiency of the sensors.

Some additional sensors with a specific purpose are described below:

- Ultrasonic sensors are extremely sensitive sensors to motion and noises. These would have very specialized uses because of the high cost. Active and passive versions are available. Active versions will be actively scanned by sending signals and waiting for the return. They have reached the level of sophistication where they can distinguish different objects and sizes and alert based on specified conditions. Passive versions will be programmed to activate on certain sounds like breaking glass.
- Vibration Motion Sensors sense small vibrations created by movement. These can be placed on the exterior of properties to detect possible intrusions. These can provide a high number of false alarms because it is difficult with just the sensor to determine what caused the vibration.
- Contact Sensors are used to monitor two contact points like a window and a windowsill. A sensor will be placed on both, and an alarm can be triggered when the sensor is separated.
- Fence Alarms can be used to determine if the gate is open, the fence has been breached, or another tampering has occurred. They can be attached to networks and monitored remotely.

Figure 1.4: Displaying a motion sensor.

Lighting

Lighting is very important because it provides accessibility to the facility during nighttime hours and can enhance safety. Having areas like parking lots, parking decks, loading docks, and entrances properly lit can help thwart crimes. There are different styles of lights that can be used to help increase the perimeter security of the operation.

- Bollards can have lighting inside them, adding an attractive border to the walkway while protecting the building from cars crashing into it.
- Wall packs are commonly mounted to the buildings' sides to illuminate small walkway areas.
- Area lights are used to illuminate a larger area than wall packs. When thinking of area lights, think streetlights.
- Floodlights are designed to “flood” the area with light. Using floodlights can lessen the number of lighting fixtures needed throughout the facility, and LED versions give long life, energy efficiency and are environmentally friendly.
- Dusk To Dawn Lights have light sensors on them to detect the ambient light. When the sun is shining, the sensor will turn the light off. When nighttime comes, and no light is triggering the sensor, the light will be turned on.
- Parking Lot and Parking Garage Lighting are necessary to help keep these areas safe. Lit areas tend to have less crime and fewer accidents.

Gated Lots

Depending on the type of business, fenced-in parking lots may be required. A security guard who checks credentials and maintains access control can be stationed at the entry and exit points of the gated lots. These lots can also be unattended and utilize a gate system that will allow access once the proper credentials are provided, like a pre-shared key, pin code or a smart card. These lots will most likely be enclosed by fencing, but natural barriers such as trees and plants may also be utilized.

Security Guards

The presence of security guards can serve as an effective deterrent and contribute to your company's safety. Security guards can be strategically placed around the building to guard and monitor specific areas of entry or where high value assets are. They can also be deployed to patrol the property identifying and investigating any suspicious activity. While some companies will oversee their own internal security team, others will choose to contract the work out to an outside company. There are numerous different tiers of protection, ranging from completely armed to completely unarmed. The organization will be responsible for determining the appropriate level of safety and protection.

Security Cameras

Security cameras installed throughout the structure can monitor critical areas and most of the building. The first areas to consider should be high traffic areas and areas with high value assets. Depending on the company's budget, these cameras come in a variety of resolutions and capabilities. Pan Zoom Tilt (PZT) cameras allow the user to move the camera in a variety of directions, allowing them to cover a large amount of ground. Some will have built-in storage, while others need to be networked and connected to the cloud. Some of them, such as those at the gates, will have audio that can be played in both directions. Cameras can also be installed on the building's exterior, where they will most likely observe the doors, pathways, and parking lot. Outdoor cameras typically include several extra features, such as the ability to withstand inclement weather, night vision, higher resolutions, and

Figure 1.5: Displaying an outdoor security camera.

Signal Blocking

When you think of signal blocking, one of the first things that may come to mind is someone jamming a cell phone. Because most countries have placed restrictions on their sale, use, and promotion, these technologies are not widely used. Because they can be used to prevent the response of emergency services and interfere with law enforcement, jammers are a risk to the general public's safety. On the other hand, organizations will implement a different type of signal blocking to increase their safety level. It is possible to protect sensitive data by utilizing both Faraday cages and geofencing.

Faraday Cage

The Faraday cage is a barrier that prevents electromagnetic radiation, such as radio waves and microwaves, from penetrating. Faraday cages are often made of conductive materials like metal sheets or wire mesh and are named after the inventor of the concept. To stop electromagnetic radiation from penetrating them completely, they will be sealed off from one end to the other.

The microwave oven is a common example of a faraday cage that may be found right in our homes. The construction prevents the microwaves from leaking out of the enclosure, which results in increased heating power for the food. To prevent PCI-DSS data from being stolen, wallets are now manufactured with RFID blocking technology like a faraday cage. For the purpose of evidence collecting, faraday bags are utilized so that electronic devices cannot be remotely controlled. The purpose of the Faraday cage is to control the transmission of signals to and from devices, which can assist in preserving data integrity.

Geofencing

Geofencing is a technology that creates an invisible barrier around a location, which then causes electronic devices to react in a certain way. The Global Positioning System (GPS) is typically utilized for this purpose; however, other forms of signals, such as cellular and Wi-Fi, may also be utilized. Once the device crosses the geofence that surrounds the area, it will be monitored for the duration that it is within the barrier. Access to network resources can be limited to only those locations within the geofence, which helps to prevent data from being stolen by unauthorized parties. There is still the possibility of insider threats.

Reflective Glass

A specific coating is applied to reflective glass, which eliminates the possibility of seeing through the glass in one direction. Because of this, unauthorized individuals are prevented from observing what is happening within the building, reducing the risk of data theft. If someone was scouting the building to break in, they couldn't easily see. Someone also would be unable to see any monitors through a window and capture information. On one side of the glass, a small layer of metal or metallic oxide has been put to the surface, which produces the illusion of a mirror on that side of the glass. The amount of light and heat that enters the structure is decreased, which in turn helps to bring down the cost of cooling the building. This is an additional benefit.

Figure 1.6: Displaying a building with reflective glass.

Data Center Camouflage

Although data centers can be built in industrial zones, where they can blend in with the other buildings, most of the time, data centers are built in remote locations with restricted access to the site. Keeping access to the data center's location private and camouflaging it with the surrounding terrain, such as a forest or mountain range, is common practice. As part of an effort to implement a defense-in-depth strategy, a secure fence and additional layers of physical security will be installed around the data center. Explosives will have difficulty cutting through thick concrete walls or creating holes in them. It is anticipated that the

Bollards

Buildings safeguarded by bollards will direct traffic and restrict the property's limits. A wide array of options are available for bollards' design, construction, and dimensions. A typical bollard will range in height from around one to two meters (three to four feet) and will be composed of concrete or steel. These bollards will most likely have a protective sleeve that is brightly colored and draws people's attention to their presence on the outside of the structure. They can prevent a car from slamming into the building while still allowing walkways for people, which increases the building's level of security.

Figure 1.7: Displaying a row of bollards.

Exterior Door Locks

Locks installed on the buildings outside doors should be of a commercial grade and aid in deterring would-be burglars from entering the structure. A few primary types of locks are typically utilized in business and institutional settings. Traditional locks and keys, keyless entry, magnetic locks, and access control systems are all available options. Individuals with the necessary credentials and who possess a smartcard, password, PIN or biometrics may be granted access using access control systems. Depending on the importance of what is being safeguarded, it may be worthwhile to invest in high-security locks that prevent lock picking and include additional sensors. As is the case with most things, these security controls can be circumvented with enough time and effort, and as such, they should not be the main layer of protection for business situations.

Exercise 2 - Interior Physical Security

After entering the building, there are various additional safety considerations to consider. Although it is critical to prepare for intruders, it is not the only danger to security that must be addressed.

In this exercise, you will learn about interior access controls and environmental controls that should be implemented.

Learning Outcomes

After completing this exercise, you should have further knowledge of:

- Interior Access Control
- Environmental Controls

Your Devices

This exercise contains supporting materials for Server+ (SK0-005).

Interior Access Control

Once inside the building, access control will be very important to help with keeping data safe. You will have simple locks and keys, but enhanced locks can be used to access and track individual movements within the organization. A frequent review of the logs should be done to note any suspicious activity.

Biometrics

Biometrics measures and analyzes a person's specific physical and behavioral attributes. A person's bodily characteristics, such as appearance, fingerprints, or voice, would be considered physical attributes. Behavior measures actions that a person takes, such as their walking motion. Biometrics have been used to identify people since the mid to late 1800s, but they are now more commonly used for access control in professional and personal environments.

Passwords can be replaced by biometrics, which is regarded as a trustworthy method of automating the authentication process. Although not very reliable, facial recognition is improving in accuracy. It's usually preferable to use something distinctive, like a fingerprint, retina scan, or iris scan. A piece of scanning equipment will collect the biometric information, and software will format it after processing it. The data will be hashed before being compared to the data kept in the table. By using this hash, which is one-way and cannot be reversed, the original data will remain secret.

Figure 2.1: Displaying a Biometric fingerprint scanner.

Radio Frequency Identification (RFID)

RFID is a term used to describe a technology in which chips or tags can be programmed with data and then used to emit a frequency that contains that data to be read by a reader for various applications. Some of the more common RFID applications are inventory tracking, near-field communication (NFC) payments, car key fobs, access control for buildings and computer systems, smartcards, identification badges, and theft protection. This technology's ease of use and low cost have contributed to its widespread adoption.

The type of RFID used can significantly impact the capabilities that can be used. Active RFID will be more expensive per RFID chip or tag, but it will have more capabilities than passive RFID. Active RFID will have a longer range, larger memory, and higher data transmission rates than passive RFID while requiring less infrastructure. Although the cost of each chip or tag associated with passive RFID will be significantly reduced, the system will require more infrastructure to function properly. Passive RFID technology would be used to track supply chains, inventories, assets, access control, and other items and prevent theft and other manufacturing-related activities

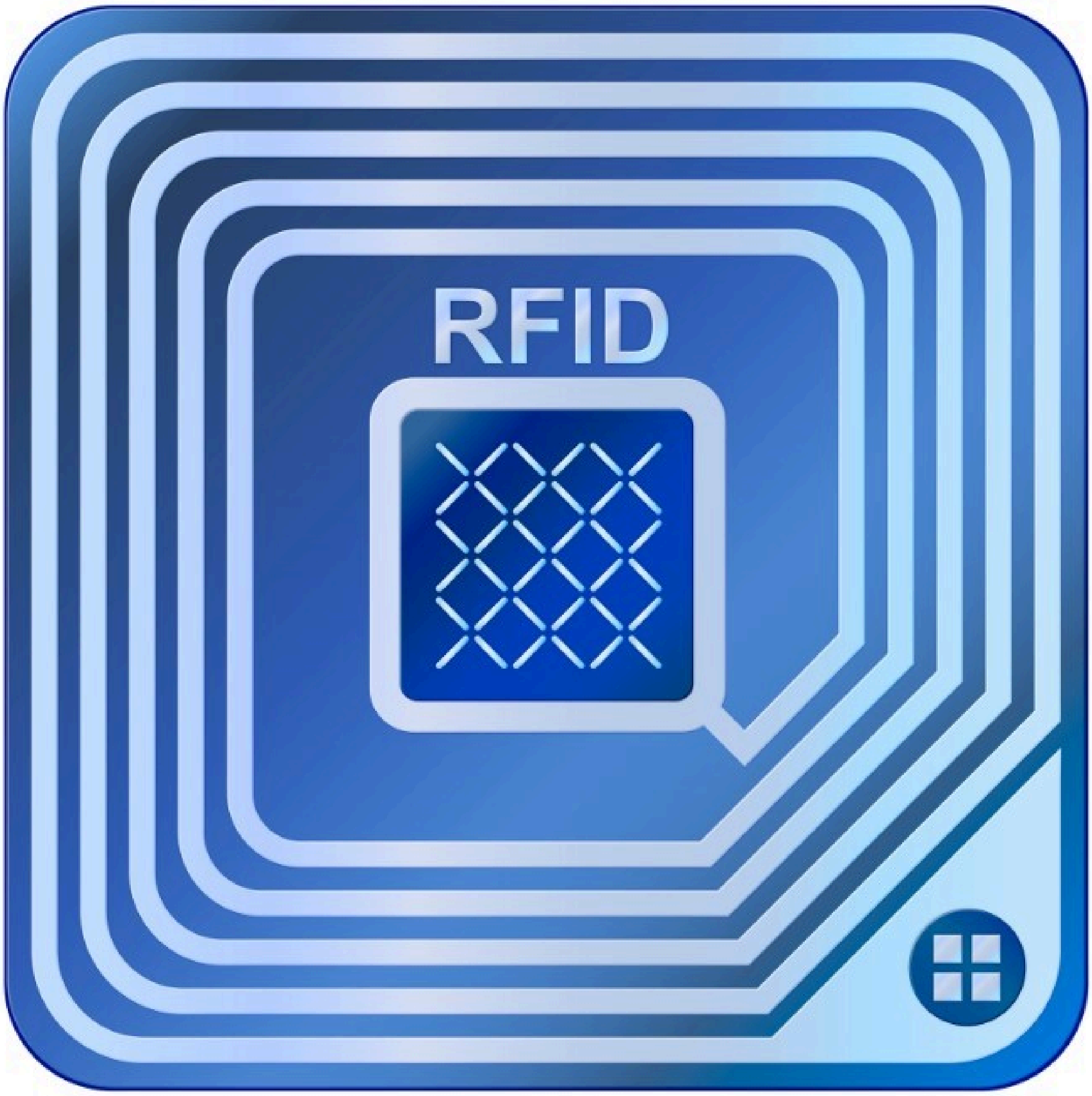

Figure 2.2: Displaying an RFID tag.

Card Readers

In order to write to and read information from a smart card, card readers are necessary. In connection with smart cards, readers are utilized in various sectors, including financial, healthcare, entertainment, travel, and security. Although some readers have built-in support for using local resources, it is more likely that they will use network resources. Contact card readers require a physical link, such as a credit card stripe or chip card. Contactless card readers will interact with the radio waves radiating from the smart card chip. A common use for these readers will be for access control. With smart cards, readers can be strategically placed around the building, allowing card holders access based on their credentials.

Figure 2.3: Displaying a door access control card reader.

Mantraps

Mantraps are used to limit physical access to one individual at a time by using a small space and a two-door system. Upon entering the small mantrap area, which should accommodate one person, the door must be closed and locked for the door in the front to be unlocked and open. The main purpose of the mantrap is to stop tailgating and piggybacking.

Figure 2.4: Displaying a Piggybacking/Tailgating attack.

Figure 2.5: Displaying a Mantrap.

Safes

Items of high value and portability, such as laptops and flash drives, will be stored in safes to prevent unauthorized access by others. Safes can incorporate various features, including a front drop for deposits, resistance to fire and water, a range of sizes, high-grade steel, and a variety of entry methods, such as a keypad or a key. Depending on their size and intended use, safes can be either portable or permanently installed, and they can be crafted from various metals. Safes of the commercial grade will typically be constructed out of heavier materials like solid steel and will include fireproofing in their design.

Environmental Controls

Fire Suppression

Fire suppression systems will almost certainly be installed in commercial settings to protect property and personnel. You are likely familiar with fire suppression systems that use water, but what about areas like our server room where water will cause as much damage as the fire? In these areas, the removal of oxygen from the atmosphere and suffocating the fire can be accomplished by utilizing pressurized gases that are not combustible. Halon, clean agent gases, carbon dioxide and inert gasses are common types of gases used.

- Clean agent gases are a type of fire extinguishant that are gaseous, electrically non-conducting, or volatile, and when they evaporate, they do not leave behind any residue. It is possible to put out Class A, B, and C fires with the use of a clean agent fire suppression system, which employs either a chemical or an inert gas to put out a fire as soon as it starts before it has the chance to spread. Clean agent fire suppression systems are safe for use in populated areas, don't require any cleanup after being discharged, and are environmentally friendly. Systems are designed and developed with the goal of protecting against water damage that is commonly caused when putting out a fire. Common Clean Agent gases are FM - 200 ™ (HFC - 227ea) and 3M ™ Novec ™ 1230 Fire Protection Fluid.
- Inert gas suppression systems use gases like argon, nitrogen, and mixtures to create fires. Argon and nitrogen are already in the air, so they don't affect the greenhouse effect and won't hurt the ozone layer. It is chemically inert, doesn't conduct electricity, is non-corrosive, and has no color, smell, or taste. They work by lowering the amount of oxygen inside the sealed off area. Inert gas is used to reduce the amount of oxygen in the air until it reaches a level where the fire is extinguished. When the inert gas is released, it spreads quickly and evenly throughout the enclosure. It takes roughly a minute to reach the concentration that is needed.
- Halon is a liquefied, compressed gas that effectively extinguishes fires. Halon is still in use, however, it is no longer manufactured. Halon is classified as a clean agent by the National Fire Protection Association since it does not conduct electricity, is deemed safe around humans, and does not leave a residue. The downside of halon is that it has a high

probability of damaging the ozone layer and contributing to global warming. More environmentally friendly options are generally available and should be considered.

- Carbon dioxide fire suppression is utilized when water cannot be used. A fire requires oxygen, fuel, and heat to continue to burn, and removing one of these three factors can extinguish a fire. When a fire is detected, carbon dioxide is released into the environment until the oxygen level is low enough to extinguish the fire. Carbon dioxide fire suppression devices are effective, however, they are hazardous to humans and animals. Carbon dioxide fire suppression systems are frequently found in vacant locations. Fire suppression systems are installed in server rooms, engine rooms, flammable storage areas, museums, data centers, and enormous industrial machines. Carbon dioxide is odorless, colorless, non-conductive, and leaves no residue.

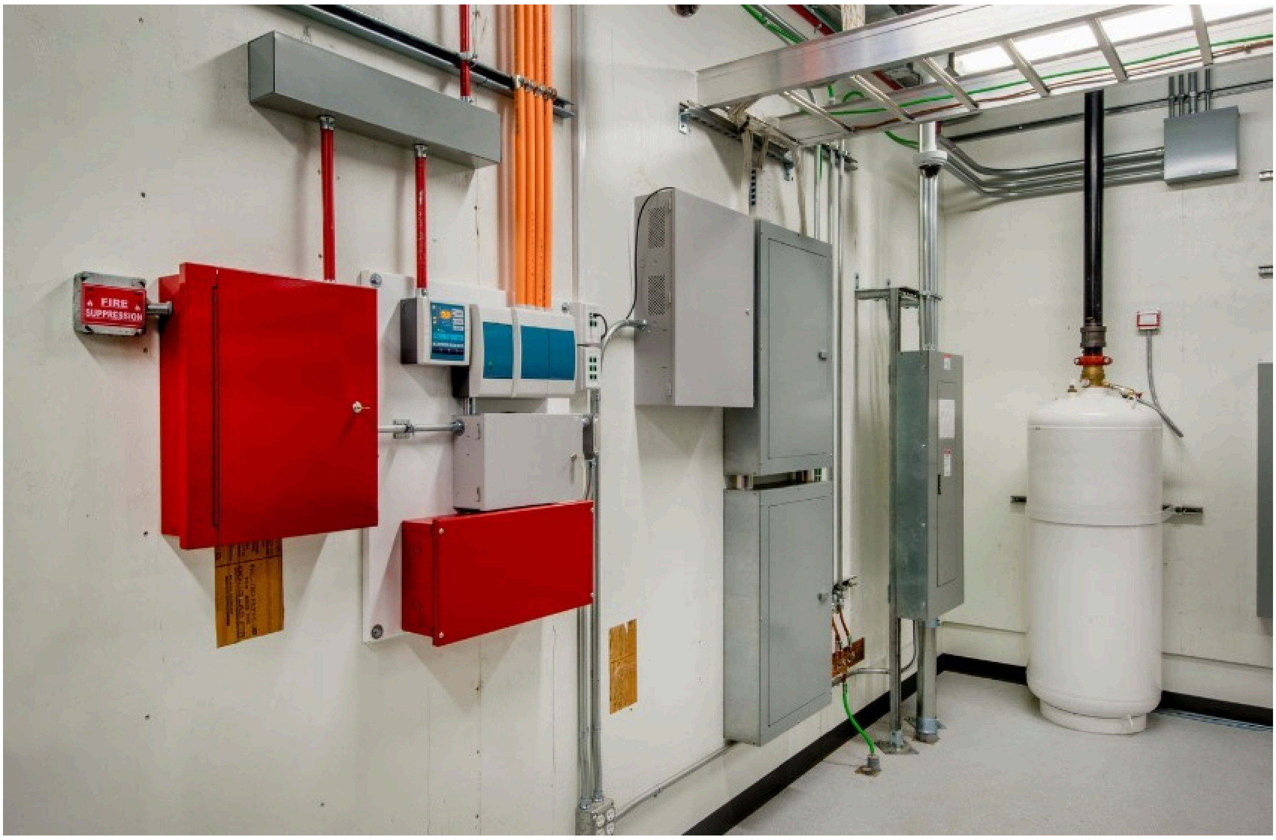

Figure 2.6: Displaying a Fire suppression system.

Heating, Ventilation, and Cooling (HVAC)

The server's surroundings must assist in keeping the server at a safe temperature. A substantial quantity of heat has gathered inside the racks, and it must be released via the ducts and the server's rear. Because SOHO server configurations will not have a lot of hardware, they may be able to get by with only pumping air conditioning into the room. When forced air is insufficient, measures such as spot coolers must be implemented, which help with both temperature and moisture in the air. Additional fans the size of a 1U server can be mounted to improve air circulation within the rack. Data centers' servers will be put on an elevated platform with air conditioning fed in through vents. The servers will be positioned so that they all face the same direction, with the intake towards the air conditioning system. The hot aisle will be positioned behind the server, and air will be taken from the area to help with temperature control and to guarantee that the environment is adequate for the servers. Redundancy is crucial, as it is with everything else, and all efforts that may be taken to assure backup cooling should be taken. The desired temperature for the room is between 68 and 76 degrees.

Humidity is an additional aspect that needs to be taken into consideration. Condensation, which can cause corrosion and damage to computer components if it forms, is more likely to occur in environments with high humidity levels. A humidity level that is too low can be just as damaging to computer equipment as one that is too high because of electrostatic discharge (ESD). This is comparable to touching a metal object while wearing socks while walking on the carpet and experiencing an electric shock. Although this has a minor effect on people, it can potentially have a devastating effect on the components of computers. The objective is to keep the room in its current state and bring the temperature down to roughly 50 percent.

These servers are extremely important to the operation of the business, so it is necessary to investigate all possibilities to ensure that they can perform as expected at all times. It is possible that additional controls, such as filters, will be required to improve the air quality in the server room.

Figure 2.7: Displaying an HVAC system for a server center.

Sensors

Sensors and monitoring systems will be installed in the server room to ensure that the temperature, humidity, smoke, and electricity levels are within acceptable parameters. These systems should include several fundamental characteristics, such as the capacity to notify managers of problems, the ability to deliver notifications by email and SMS, and an intuitive user interface.

Some of the monitoring systems will be software, such as a website or an application that can be downloaded to a mobile device, computer, or tablet; other systems will be hardware, such as a standalone monitoring device. Both versions will work toward providing the user with a single dashboard where they can view all of the notifications related to the connected devices in their environment. Some may even be able to incorporate notifications into their security system if the rack doors are opened. It's possible that you'll have to pay a monthly subscription charge depending on the system.

Review

Well done, you have completed the **Securing a Physical Server Infrastructure** Practice Lab.

Summary

You completed the following exercises:

- Exercise 1 - Exterior Physical Security
- Exercise 2 - Interior Physical Security

You should now have further knowledge of:

- Fences & Security Walls
- Motion Sensors
- Lighting
- Gated Lots
- Security Guards
- Security Cameras
- Signal Blocking
- Reflective Glass
- Data Center Camouflage

- Bollards
- Exterior Door Locks
- Interior Access Control
- Environmental Controls

Feedback

Shutdown all virtual machines used in this lab. Alternatively, you can log out of the lab platform.